01 / PRODUCT INTENT

Hot-cup *protein beverage.*

An enjoyable everyday cup, prepared with **hot water or hot milk.**

Nurture by TRAYN Nutrition is a proposed dairy-based powder with protein and selected micronutrients. The intended occasion is a daily beverage that customers enjoy enough to choose in place of a habitual tea or coffee cup.



Generated packaging concept
Artwork and claims unapproved

Review requested
Please assess feasibility, recommend a formulation direction and identify the tests and changes needed before committing to a product specification.

Development parameters

Preparation goal	**Both hot water and hot milk.** Spoon preparation should be practical in an ordinary kitchen. Validate each liquid separately.
Working trial range	Explore **10-15 g protein supplied by the powder** per serving, initially around **180-200 mL prepared beverage**. These are trial parameters, not an approved dose or claim.
Audience for review	Founder's initial audience: women **18-55**. Assess broader adult suitability, potentially including 55+. A **25-45** marketing focus is a hypothesis, not a nutritional cutoff.
Sensory goal	Aromatic, smooth and lightly creamy; enjoyable through the last sip. Evaluate foam, graininess, drying, aftertaste and sediment. Initial flavour ideas: cocoa-vanilla or cardamom-vanilla.
Use and positioning	One daily serving is the proposed design basis. Assess plausible multiple-cup use. Caffeine choice is open; this is not a meal-replacement or deficiency-treatment brief.

Inputs still to confirm: target price per powder serving, serving weight, pack format, manufacturing scale and caffeine position. India is the assumed launch market. Final nutrient doses, preparation instructions and shelf life remain open.

Assess dairy-allergen/lactose implications and intended use explicitly. Pregnancy and breastfeeding require a separate formulation assessment.

02 / FORMULATION FOR ASSESSMENT

Protein system and fortification

Proposed first route
Evaluate milk protein concentrate (MPC) with added whey, plus a controlled milk-powder contribution for flavour and body. Assess a small fat adjustment only if it improves the complete cup.

This architecture is open to your recommendation. Ingredient grades, total casein:whey balance, processing and proportions are not fixed. Milk solids and concentrated milk protein are not interchangeable.

Compare	Question to resolve
A Whey-led base	Whey with some skim milk powder: can the simpler system meet both preparation and sensory goals?
B MPC + whey	A milk-protein/whey system with less skim milk powder: does it improve the balance of hydration, stability and body?
C Creaminess	Modify the better base with a small fat contribution: is the benefit worthwhile without excessive richness in milk?

Compare at matched protein contribution and similar sweetness. **Hydration and hot-cup stability are separate tasks.** Research supports investigating casein/whey systems and agglomeration, but does not validate this proposed household use. [1, 2]

Micronutrients: shortlist, not specification

Candidate	Evaluation requested
B12 + vitamin D	Assess nutritional rationale, permitted forms, amounts and retention. No universal deficiency or outcome benefit is assumed.
Calcium	Count the base and preparation-milk contribution first. Evaluate any addition for nutritional need, taste and physical stability.
Iron	Make a separate life-stage and formulation decision. Advise whether inclusion is justified in a broad adult product or a more specific direction. [3]

Nutrition design: propose a defined powder dose for both liquids where feasible; calculate each prepared cup separately. Assess total intake, other supplements, tolerance, premix uniformity and nutrient retention. Use applicable Indian references and the confirmed FSSAI category. [4, 5]

03 / YOUR EVALUATION

Questions for your review

Please challenge the assumptions as well as the proposed route. We would value a practical recommendation on the following six decisions.

01

Protein and serving
Which protein system, ingredient grades, protein contribution and powder serving weight would you recommend? Please explain the tradeoffs.

02

Hot preparation
Can both liquids be supported reliably? Specify a feasible mixing method, temperature window, liquid quantities and standing period to validate.

03

Audience and nutrients
What adult age/life-stage positioning and micronutrient package are defensible? Assess iron separately and identify any intended-use limitations.

04

Validation and packaging
What bench, sensory, analytical, microbiological, nutrient-uniformity, packaging and shelf-life work is needed before launch?

05

Manufacturing and cost
What instantisation/process capabilities, supplier data and commercial inputs are needed? Identify the major cost and scale constraints.

06

Proceed, revise or pivot
What should change if protein level, dual-liquid performance, taste and price conflict? Identify the decision gates before further investment.

Requested response
Your preferred route and reasons; changes to the brief; priority risks; an initial prototype and validation plan; and the commercial information needed from us.

Selected references

[1] Singh et al. (2019). Casein/whey UHT stability. Food Research International.

[2] Seo (2022). Agglomeration of whey-fortified milk powder.

[3] NIH Office of Dietary Supplements. Iron: life-stage background.

[4] ICMR-NIN. Nutrient requirements and RDA/EAR publications.

[5] FSSAI. Category framework (check applicability and current rules).

FSSAI advertising and claims: accompanying regulations and amendments.

References inform the questions; they do not establish a Nurture formula or approval. The wider discussion, source notes and open decisions are preserved in the companion Markdown.